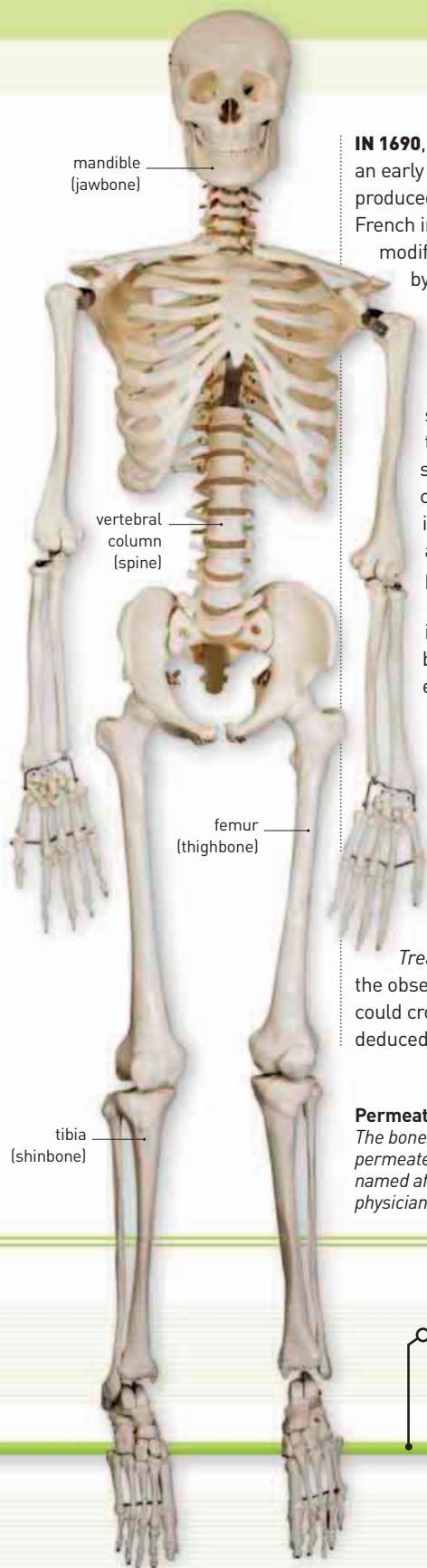


“ONE MAY CONCEIVE **LIGHT** TO **SPREAD** SUCCESSIVELY, BY **SPHERICAL WAVES.**”

Christiaan Huygens, Dutch physicist, from *Treatise on Light*, 1690



IN 1690, a decade after making an early pressure cooker that produced high-pressure steam, French inventor **Denis Papin** modified his original design by incorporating a piston, producing the **first working “atmospheric” engine**. Boiling water in a cylinder created steam, which pushed the piston up; as the steam condensed, it created a vacuum in the cylinder and atmospheric pressure plunged the piston back down. This invention marked the beginning of steam engine development. Papin received advice on his designs from Dutch astronomer **Christiaan Huygens** (1629–95) who, also in 1690, made a significant contribution to other areas of knowledge with his *Treatise on Light*. Based on the observation that light beams could cross without bouncing, he deduced that **light is composed**

Permeated bones

The bones of the body are permeated by tiny organic channels, named after Clopton Havers, the physician who discovered them.

of waves—supporting a theory proposed by French philosopher René Descartes in the 1630s and English inventor Robert Hooke in the 1660s. But Isaac Newton’s idea that light was made from particles (see 1675–84) was to predominate over the wave theory for more than 100 years.

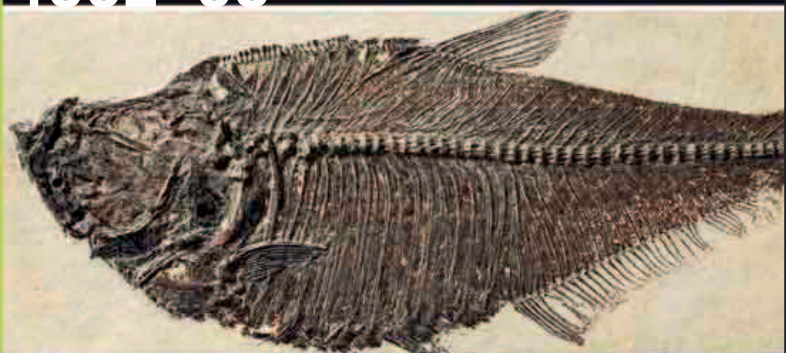
Clopton Havers (1657–1702), an English physician, was the first to study the **detailed anatomy of bones**—including marrow and cartilage. He published his results in 1691, describing the microscopic pores and cavities running through a bone’s structure. Havers surmised that they carried oil, but it is now known that these so-called **Haversian canals** contain blood and lymphatic vessels, and provide bone cells with oxygen and nourishment.

70 PERCENT THE AMOUNT OF BONE MADE OF NONLIVING MINERAL

1690 French inventor Denis Papin develops a piston steam engine

1690 Dutch astronomer Christiaan Huygens describes a wave theory in his *Treatise on Light*

1691 English physician Clopton Havers publishes a treatise on bone anatomy



The location of fossil fish and other marine creatures far inland gave rise to conflicting theories among naturalists and theologians.



John Ray
Philosopher and theologian, John Ray is also regarded as the founding father of English natural history.

have created the specific layers that could be observed in geological deposits. He suggested that in an ancient world covered by sea, land rose by volcanic activity—which would explain the occurrence of fossilized marine animals on land. However, Ray’s theological leanings meant that he was reluctant to take the view that divinely created species could become extinct. He proposed

“**NEVER... DID I EXPECT TO PRODUCE A HISTORY OF QUADRUPEDS.**”

John Ray, from *Synopsis of Quadruped Animals*, 1693

that organisms so far known only as fossils would one day be found living in remote areas.

In 1693, Ray published one of his most important works of zoology, *Synopsis of Quadruped Animals and Serpents*. Based on anatomical features, it provided

IN 1692, SCOTTISH PHYSICIAN JOHN ARBUTHNOT (1667–1735) published *Laws of Chance*. This was a translation of Christiaan Huygen’s 1657 classic work on **probability theory**, and the first publication in English devoted to the subject.

English naturalist **John Ray** had written extensively on the diversity of plants since the 1660s, but by the 1690s he was also active in paleontology (the study of fossil organisms) and zoology. His accurate descriptions of fossils supported the idea that they were the remains of once-living species. Ray also **tried to explain the locations of fossils**. A popular concept was that the Biblical flood had been responsible for the forming of fossils, but Ray saw that a deluge would not

1692 Scottish physician John Arbuthnot publishes the first English work on probability theory